EXAM SHIELD

Cyber Security Question Bank

Security principles, threats, cryptography and defence.

13 questions: 8 multiple choice, 3 fill in the gap and 2 written. Total 20 marks.
Edition: with answers (staff copy, ready to import into Exam Shield).

How this paper is laid out

Questions are numbered in one sequence. Multiple-choice options are lettered with a full stop; each multiple-choice question is worth one mark unless stated.

Fill-in-the-gap questions show each blank as a line; they earn one mark per blank, shown in square brackets, and are marked automatically.

Written questions show their marks in square brackets and are marked by an examiner.

To use this paper in Exam Shield, open an exam, choose Questions and then Import from PDF, and upload this file. Review the preview and import.

Section 1: Cyber Security

Security principles, threats, cryptography and defence.

1. What do the letters of the CIA triad stand for?

A. Cipher, Identity, Access
B. Control, Inspection, Audit
C. Compliance, Integrity, Authentication
D. Confidentiality, Integrity, Availability

Answer: D

2. What is phishing?

A. Guessing passwords by trying every combination
B. Tricking people into revealing information through fake messages
C. Capturing packets on a network
D. Flooding a server with requests

Answer: B

3. Which of these is an asymmetric (public-key) algorithm?

A. AES
B. DES
C. 3DES
D. RSA

Answer: D

4. What is the most effective defence against SQL injection?

A. Longer passwords
B. Parameterised queries
C. Using HTTPS
D. Installing a firewall

Answer: B

5. What does multi-factor authentication require?

A. Two or more independent proofs of identity
B. Two or more administrator accounts
C. Two or more firewalls
D. Two or more backups

Answer: A

6. Which property describes a cryptographic hash function?

A. It is one-way and cannot practically be reversed
B. It compresses files losslessly
C. It can be decrypted with the right key
D. It is the same as Base64 encoding

Answer: A

7. A DDoS attack mainly threatens which security goal?

A. Confidentiality
B. Integrity
C. Non-repudiation
D. Availability

Answer: D

8. What does the principle of least privilege mean?

A. Every user is an administrator
B. All users have read-only access to everything
C. Passwords are not required
D. Users get only the access they need to do their work

Answer: D

9. A _____ filters network traffic according to a set of security rules. [1 mark]

Answer: firewall

10. Malware that encrypts a victim's files and demands payment for the key is called _____. [1 mark]

Answer: ransomware

11. Simulating a real attack on a system, with the owner's permission, to find weaknesses is called penetration _____. [1 mark]

Answer: testing

12. Explain the difference between symmetric and asymmetric encryption, with an example of each. [4 marks]

13. Describe the main phases of an incident response plan. [5 marks]